ANALYSIS OF THE DEPENDENCE BETWEEN ASSETS

USING MULTIVARIATE LÉVY MEASURES AND LÉVY COPULAS

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Overview

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2. Data regimes
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4. Low-frequency methods
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6. Lévy copula approach
7. Dependence concepts via Lévy-Clayton copula

Lévy processes

Definition (Lévy process $(X_t)_{t\geqslant 0}$)

Bivariate stochastic process $(X_t)_{t\geqslant 0}$ is a Lévy process if

- $X_0 = 0$ a.s.;
- independent and stationary increments;
- cádlág trajectories, $\Delta X_t := X_t - X_{t-}$.

Lévy-Ito decomposition:

$$\mathbf{X}_t = \underbrace{ct + \mathbf{W}_t}_{\text{Gaussian part}} + \underbrace{\int_0^t \int_{0<\|x\|<1} x(\mu - \bar{\mu})(dx, ds)}_{\text{Small jumps, any intensity}} + \underbrace{\int_0^t \int_{\|x\|\geqslant 1} x\mu(dx, ds)}_{\text{Big jumps, finite intensity}},$$

where $c \in \mathbb{R}^2$, $(\mathbf{W}_t)_{t\in\mathbb{R}_+}$ - bivariate Brownian motion $(0, \Sigma)$, μ and $\bar{\mu}$ are the bivariate jump measure and its compensator. Lévy measure ν satisfies $\bar{\mu}(dx, ds) = ds\ \nu(dx)$.

$$\nu(A) = \mathbb{E}\{\#\text{ jumps} : \Delta X_t \in A,\ t \in [0,1]\} \quad \forall A \in \mathcal{B}(\mathbb{R}\setminus\{0\})^2$$

High and low frequency data regimes

High-frequency data regime

- $\delta = T/n \to 0$, $n \to \infty$
- ticks, 1 sec
- moments and magnitudes of jumps can be observed

Low-frequency data regime

- $\delta = T/n = const$
- 1 minute, 1 hour
- moments and magnitudes of jumps are latent variables

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High-frequency method: method of sieves

Idea:

To approximate Lévy density s(x):

$$\nu(A) = \int_A s(x)\,dx \qquad \forall A \in \mathcal{B}\left((\mathbb{R} \setminus \{0\})^2\right)$$

over a window $D = ([a, b] \setminus [-\varepsilon, \varepsilon])^2$ using a sieve- a finite-dimensional space consisting of the orthonormal functions $\varphi_j : D \to \mathbb{R}$:

$$\mathcal{S} := \{\beta_1\varphi_1 + \ldots + \beta_d\varphi_d, \beta_1, \ldots \beta_d \in \mathbb{R}\}.$$

Best approximation of $s(x)$ using functions from $\mathcal{S}$- orthogonal projection $s(x)$ on $\mathcal{S}$:

$$s^{\perp}(x) = \sum_{i=1}^{d} \beta(\varphi_i)\varphi_i(x), \qquad \beta(\varphi_i) = \langle s, \varphi_i \rangle_D = \int_D s(x)\varphi_i(x)dx.$$

High-frequency method: method of sieves

- Small time asymptotics: $\exists k > 0, t_0 > 0$:

$$\sup_{y \in D} \left| \frac{1}{\delta} \mathbb{P}[X_\delta \geqslant y] - \nu([y, \infty)) \right| < k\delta \qquad \forall 0 < \delta < t_0$$

- Under high-frequency regime $\delta \to 0, \ n \to \infty$:

$$\hat{\beta}(\varphi_i) := \frac{1}{n\delta} \sum_{k=1}^{n} \varphi_i(\underbrace{\Delta X_{k\delta}^{(1)}}_{i.i.d}, \underbrace{\Delta X_{k\delta}^{(2)}}_{i.i.d}) \xrightarrow[n \to \infty]{LLN} \frac{1}{\delta} \mathbb{E}\left[\varphi_i(X_\delta^{(1)}, X_\delta^{(2)})\right] =$$

$$= \frac{1}{\delta} \int_D \varphi_i(x) P_\delta(dx) \xrightarrow[\delta \to 0]{\text{Small time asymptotics}} \int_D \varphi_i(x) s(x) dx = \beta(\varphi_i)$$

- The estimator $\hat{s}(x)$ for $s^\perp(x)$ is

$$\hat{s}^\perp(x) = \sum_{i=1}^{d} \hat{\beta}(\varphi_i) \varphi_i(x).$$

Low-frequency method: spectral approach

Here $\varphi_\delta(u) = \mathbb{E}\exp\{iuX\} = \exp\{\delta\Psi(u)\}$ Using Laplacian and Fourier transformation $\mathcal{F}$:

$$\Delta\Psi = -\operatorname{tr}(\Sigma) - \int_{\mathbb{R}^2\setminus\{(0,0)\}} \|x\|^2 e^{i\langle u,x\rangle} s(x)\,\mathrm{d}x = -\operatorname{tr}(\Sigma) - \mathcal{F}[\|x\|^2 s]$$

Empirically, Laplacian can be obtained as

$$\widehat{\Delta\Psi} = \frac{\hat\varphi_\delta\,\Delta\hat\varphi_\delta - \sum_{j=1}^2 (\partial_{u_j}\hat\varphi_\delta)^2}{\delta\,\hat\varphi_\delta^2}\mathbf{1}_{\{|\hat\varphi_\delta|\geq T^{-1/2}\}},$$

where $Y_k = X_{k\delta} - X_{(k-1)\delta}$ and

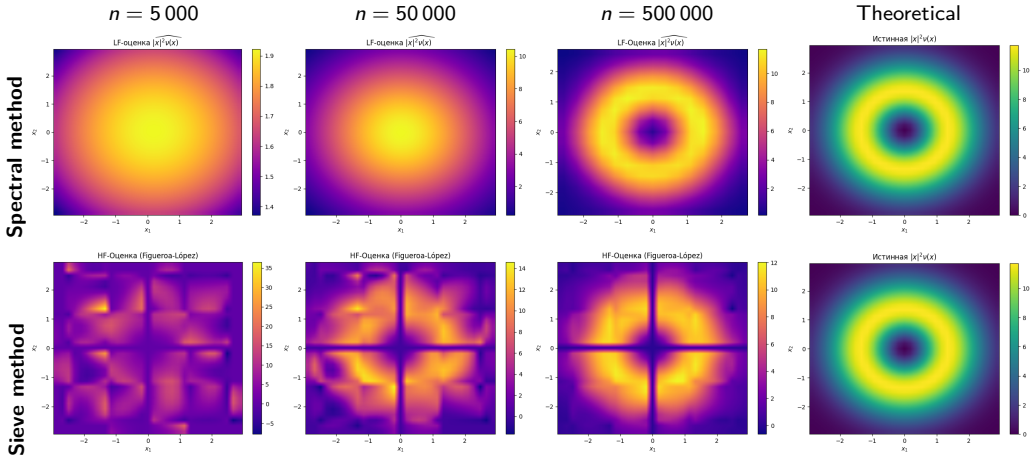
$$\hat\varphi_\delta(u) = \frac{1}{n}\sum_{k=1}^n e^{i\langle u,Y_k\rangle}, \quad \partial_{u_j}\hat\varphi_\delta(u) = \frac{i}{n}\sum_{k=1}^n \Delta Y_k^j e^{i\langle u,Y_k\rangle}, \quad \Delta\hat\varphi_\delta(u) = -\frac{1}{n}\sum_{k=1}^n \|\Delta Y_k\|^2 e^{i\langle u,Y_k\rangle}$$

are empirical characteristic function its derivative and Laplacian.

Choosing a kernel K of order p such that $\int K = 1$ and $\operatorname{supp}\mathcal{F}K$ is compact with a bandwidth $h = h_n \to 0$,

$$\widehat{\|x\|^2 s} = -\mathcal{F}^{-1}\Big[\mathcal{F}[K_h]\,\big(\widehat{\Delta\Psi} + \widehat{\operatorname{tr}(\Sigma)}\big)\Big]$$

Simulation study



Dependence concepts via Lévy measure

Dependence (complete)

Complete dependence of the jump components of $(\mathbf{X}_t)_{t\geqslant 0}$ implies that

$$\exists\, \mathcal{I} \subset (\mathbb{R}\setminus\{0\})^2 - \text{an increasing subset}$$

such that the support of the Lévy measure $\nu(x)$ belongs to $\mathcal{I}$.

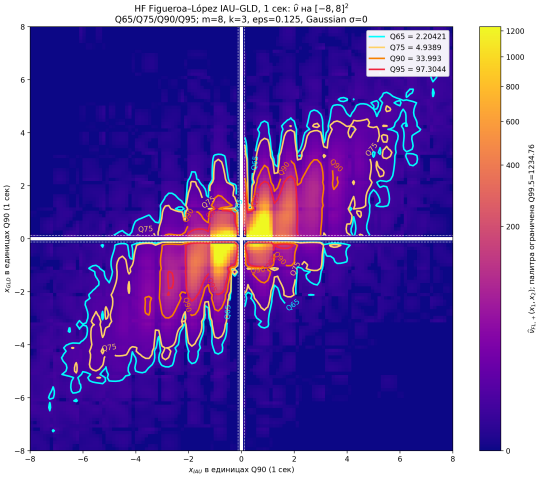
Independence

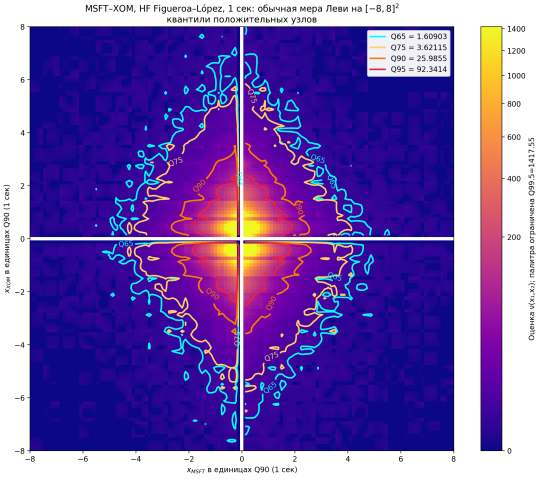
Independence of the jump components of $(\mathbf{X}_t)_{t\geqslant 0}$ implies that Lévy measure ν has the following form:

$$\nu(A) = \nu_1(A_1) + \nu_2(A_2),$$

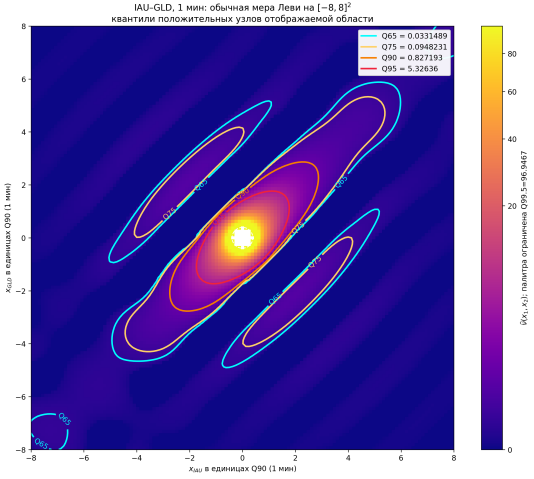
where $A_1 = \{x_1 : (x_1, 0) \in A\},\ A_2 = \{x_2 : (0, x_2) \in A\}$ and ν_1, ν_2 are Lévy measures of $X_t^{(1)}, X_t^{(2)}$ respectively.

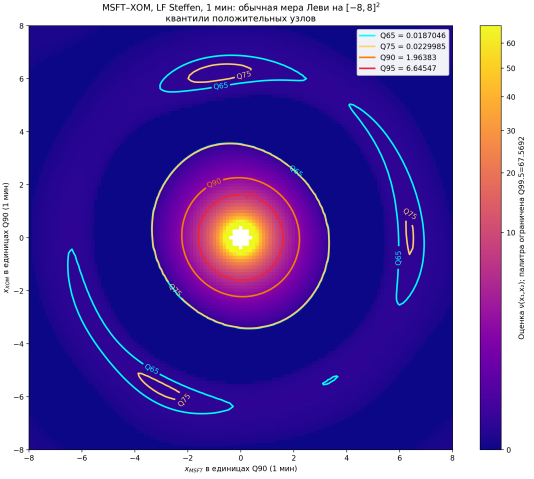
Lévy measure estimation under high-frequency regime





Lévy measure estimation under low-frequency regime





Lévy copula

Definition (Bivariate Lévy copula)

Bivariate Lévy copula is a 2-increasing grounded function $\mathbf{F}(x, y) : [-\infty, \infty]^2 \to [-\infty, \infty]$ which has uniform margins: $\mathbf{F}(x, +\infty) - \mathbf{F}(x, -\infty) = \mathbf{F}(+\infty, x) - \mathbf{F}(-\infty, x) = x$.

Key property:

$$\nu(x_1, x_2) = \frac{\partial^2 \mathbf{F}(y_1, y_2)}{\partial y_1 \partial y_2}\Bigg|_{y_1 = U_1(x_1), y_2 = U_2(x_2)} \nu_1(x_1)\nu_2(x_2)$$

Example: Lévy-Clayton copula with two dependence parameters ($\theta^+ > 0, \theta^- < 0$)

$$\mathbf{F}_{(\theta^+,\theta^-)}(y_1, y_2) = \left(c_1|y_1|^{-\theta^+} + c_2|y_2|^{-\theta^+}\right)^{-1/\theta^+} \cdot \mathbf{1}_{\{y_1 \cdot y_2 \geqslant 0\}} - \\ - \left((1-c_1)|y_1|^{\ \theta^-} + (1-c_2)|y_2|^{\ \theta^-}\right)^{1/\theta^-} \cdot \mathbf{1}_{\{y_1 \cdot y_2 \leqslant 0\}},$$

where $c_1, c_2 \in [0, 1]$.

Empirical copula

Bivariate tail integral is a 2-increasing, left-continuous function $U : (\mathbb{R} \setminus \{0\})^2 \to \mathbb{R}$

$$U(x_1, x_2) := \nu\left(\Upsilon(x_1) \times \Upsilon(x_2)\right) \prod_{i=1}^{2} \operatorname{sign}(x_i), \qquad \Upsilon(x) = \begin{cases} [x, \infty], & x \geqslant 0, \\ [-\infty, x], & x < 0. \end{cases}$$

Key result (Consequence from Sklar's theorem)

Let $U(x_1, x_2)$ be a bivariate tail-integral with U_1, U_2 margins. Then, a Lévy copula $\mathbf{F}$ can be defined separately on each quadrant by

$$\mathbf{F}(y_1, y_2) = U(U_1^{-1}(y_1), U_2^{-1}(y_2)) \qquad \forall (y_1, y_2) \in [-\infty, \infty]^2.$$

Empirical copula is constructed as

$$\widehat{\mathbf{F}}(y_1, y_2) = \widehat{U}(\widehat{U}_1^{-1}(y_1), \widehat{U}_2^{-1}(y_2)),$$

$$\widehat{U}(x_1, x_2) = \left(\prod_{i=1}^{2} \operatorname{sign}(x_i) \right) \cdot \frac{1}{n\delta} \sum_{k=1}^{n} \mathbf{1}_{\{\Delta X_{k\delta}^{(1)} \in \Upsilon(x_1), \;\; \Delta X_{k\delta}^{(2)} \in \Upsilon(x_2)\}}.$$

Dependence concepts via Lévy copula

Independence

Independence of the jump components of $(\mathbf{X}_t)_{t\geqslant 0}$ implies that the form of the copula for such a process is

$$\mathbf{F}_{ind}(y_1, y_2) = y_1 \cdot y_2$$

Dependence

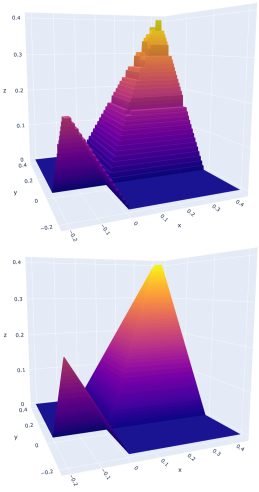
Examples of the Lévy copulas that decribe complete dependence are

$$\mathbf{F}_p(y_1, y_2) = \min(|y_1|, |y_2|)\mathbf{1}_{\{y_1 y_2 \geqslant 0\}} \qquad \mathbf{F}_n(y_1, y_2) = -\min(|y_1|, |y_2|)\mathbf{1}_{\{y_1 y_2 \leqslant 0\}}$$

Quantitative measure of dependence

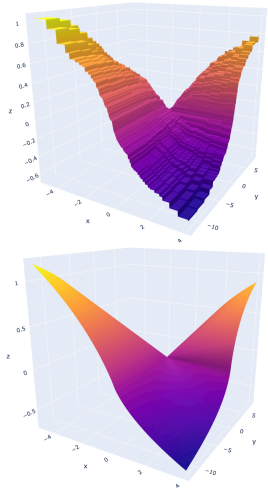
$$\theta_{adj} = \frac{\theta^+ + \theta^-}{\theta^+ - \theta^-} \in (-1, 1)$$

Lévy copula estimation results



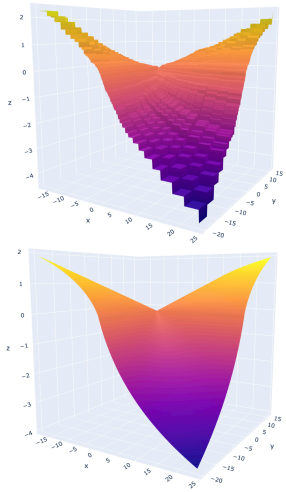
GLD-IAU, $\theta_{adj} = 0.997$

similar to $z = \min(|x|, |y|)$



AAPL-IBM, $\theta_{adj} = 0.16$

similar to $z = x \cdot y$



NVDA-PEP, $\theta_{adj} = -0.46$

counter-directed jumps

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